

Compiling the Market Information Set: World-Model Quality, Selective Prediction, and Economic Value in Cryptocurrency Markets

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Canonical TeX with displayed equations: pdf/sci/main_jf_rfs.tex. Chinese full manuscript: pdf/cn/main_cn_jf.md. Theory source: pdf/original/. EvoQuant is the laboratory, not the theory source.

Abstract

Conditional asset pricing treats the investor’s information set as primitive. In cryptocurrency markets that premise fails: evidence arrives asynchronously and is intermittently unavailable. This paper develops a finance-native theory of *information-set compilation*. We define epistemic observations, asynchronous reconstruction bounds, a compilation operator from raw to AI-visible filtrations, WMI, and regime-conditional ACWMI; and we formalize ECP, MIG, an availability-shock identification DAG, Bayesian abstention, and explanation metrics EAR/UCR/EV.

Empirically, we populate a real multi-band archive and construct a 400-day PIT panel (4000 asset-days) aligned to Yahoo returns. Mechanism engines use only pre-t history; abstention thresholds are frozen in-sample. Out of sample, thick real PIT worlds dominate exchange-only thin worlds (CE 0.474 vs -0.011). Leave-one-band-out on durable bands shows large CE losses from dropping macro (-0.534), alternative (-0.526), or exchange (-0.339). An IS-frozen ACWMI gate remains implementable (Sharpe 0.901, CE 0.199, abstain 29.7%) but does not dominate ungated thick signals on CE. The production WMI <0.2 rule abstains 100% on the sparse archive. Contribution: compilation theory with displayed primitives plus a reproducible PIT identification protocol. Limitations are stated as the agenda for a final JF/RFS submission.

Keywords: information set; selective prediction; cryptocurrency; point-in-time; measurement error; world models. **JEL:** G12, G14, C58, C55

1. Introduction

Modern asset pricing conditions on an information set I_t . The literature’s discipline has focused on how I_t is used—for discount factors, return prediction, and machine-learning pricing kernels—not on how I_t is compiled from asynchronous, quality-heterogeneous evidence. In cryptocurrency markets, compilation is first-order: venue fragmentation, perpetual leverage, options walls, unlocks, on-chain flows, and macro liquidity can map the same price path into incompatible states.

This paper makes three contributions. **Theory:** Regime-Conditional Adaptive World Models (RCA-WM) with epistemic observations, lag bounds, compilation operator Π_t , WMI/ACWMI, ECP, MIG, identification DAG, Bayesian abstention, and EAR/UCR/EV. **Laboratory:** EvoQuant as a reproducible measurement instrument. **Identification:** a real multi-band PIT archive with frozen thresholds and leave-one-band-out economic value.

2. Related literature

Information sets in asset pricing. Classic and modern frameworks treat I_t as given (Cochrane 2005; Fama–French 1993; Gu–Kelly–Xiu 2020; Kelly–Pruitt–Su 2019; Nagel 2021; Harvey–Liu–Zhu 2016). Machine-learning asset pricing expands the feature span inside I_t but typically assumes synchronized panels. We study the prior step: compilation.

Cryptocurrency market structure. Fragmentation and distinctive risk factors (Makarov–Schoar 2020; Liu–Tsyvinski–Wu 2022) motivate multi-band worlds rather than price-only predictors.

Selective prediction and measurement timing. When the world is thin or dishonest, abstention can be optimal. Macro vintages and crypto freshness/missingness make measurement timing first-class.

3. Theory

This section restores the formal World-Model-First / RCA-WM apparatus. Empirics instantiate these objects; they do not replace them.

3.1 World-model quality: breadth, stability, honesty

Let K be the number of critical evidence bands and $a_{k,t} \in \{0,1\}$ an AI-usable indicator. Breadth, stability, and honesty are:

$$B_t = (1/K) \sum_k a_{k,t}, \quad U_t = \exp(-\sum_j \omega_j d_{j,t}), \quad H_t = 1 - (1/J) \sum_j m_{j,t}.$$

The production baseline World Model Index is the product form

$$WMI_t = B_t \times U_t \times H_t.$$

Hierarchical breadth and continuous honesty used in the PIT panel:

$$B_t^{\text{hier}} = 0.25 B_t^{\text{dom}} + 0.35 B_t^{\text{band}} + 0.40 B_t^{\text{asset}},$$

$$H_t^{\text{cont}} = \exp(-2c_t) \max(0, 1 - 0.5(1 - e_t)),$$

where e_t is the ready-share and c_t a contamination share.

3.2 Epistemic observations

An AI-consumable market observation is not a bare scalar:

$$O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t}),$$

with value, latest-available time, quality, main-view gate, and semantic role. Numerically identical x with different (τ, q, g, r) are different world-model objects.

3.3 Asynchronous state space and lag error

Latent states evolve as $S_{t+1} = F(S_t, \eta_{t+1})$. Source j observes a lagged map $X_{j,t}^{\text{obs}} = h_j(S_{t-\tau}) + v_{j,t}$. If h_j is Lipschitz, reconstruction error admits a bound with delay, noise, and missingness terms:

$$\|S_t - S_t^{\text{obs}}\| \leq C_1 \sum_j \omega_j \|x_{j,t}\| + C_2 \sum_j \omega_j \|v_{j,t}\| + C_3 \sum_j \omega_j (1 - z_{j,t}).$$

Latest-snapshot, freshness/TTL, and readiness gates target these three terms.

3.4 Filters and the compilation operator

$$F_t^{\text{raw}} = \sigma(\{X_{j,t}^{\text{obs}}\}), \quad F_t^{\text{AI}} = \sigma(W_t^{\text{AI}} D_t), \quad \Pi_t = B_t M_t A_t, \quad W_t^{\text{AI}} = \Pi_t (F_t^{\text{raw}}).$$

Proposition (compilation $\neq$ feature expansion). Enlarging F_t^{raw} without a well-defined Π need not enlarge decision-relevant F_t^{AI} : ungated, stale, or role-incoherent evidence can expand raw span while shrinking usable world quality via H_t and U_t .

3.5 ECP, MIG, and causal DAG

$$ECP_t = 1\{\text{conf}_t > c\} 1\{WMI_t < w\}.$$

Thick versus thin information sets differ by conditional mutual information; band k 's marginal information gain on task m is $MIG_{k,t}^{(m)} = I(R_t^{(m)}; E_{k,t} | I_t^{(-k)})$.

Identification DAG: $O_t \rightarrow W_t \rightarrow A_t$, with market complexity M_t and latent configuration C_t as confounders. Availability shocks O_t are the preferred quasi-exogenous lever.

3.6 Bayesian abstention

With action set $A = \{\text{bullish, bearish, neutral, abstain}\}$,

$$a_t^* = \arg \min_a E[\mathbf{L}(a, R_t) | W_t].$$

Abstain when every non-abstain action has expected loss above a world-dependent cost $c_{\text{abs}}(W_t)$.

3.7 ACWMI and explanation metrics

$$\text{ACWMI}_t = \exp(\sum_i \gamma_i(r_t) \log x_{i,t} / \sum_i \gamma_i), \text{ with } x = (B^{\text{hier}}, U, H^{\text{cont}}, S, C).$$

$\text{EAR}_t = (\# \text{ evidence-bound claims}) / (\# \text{ claims})$, $\text{UCR}_t = 1 - \text{EAR}_t$, $\text{EV}_t = d(\Phi_t, \Phi_{t-1}) / (1 + d(W_t, W_{t-1}))$.
Thin and thick worlds also differ in explanation sets Φ_t .

4. The EvoQuant laboratory as measurement instrument

EvoQuant supplies multi-band collectors, readiness scoring, BandPITService / time_slice PIT reconstruction, availability-shock queries over collection_runs, and configurable WMI/ACWMI thresholds (WORLD_MODEL_INDEX_MODE, WMI_ABSTAIN_THRESHOLD, ACWMI_ABSTAIN_THRESHOLD). Epistemological order: theory first, laboratory second.

5. Data and real multi-band PIT archive

5.1 Populated archive

Exchange (OKX): ~9373 klines/merged bars, daily 2025-06-30 $\rightarrow$ 2026-08-03; ~2448 funding rows. Macro: 81013 vintaged points. Alternative: 88761 points. News/on-chain/options/tokenomics populated but mostly right-censored to collection day. Analytics snapshots remain sparse historically; multi-band PIT uses raw history tables via `build_pit_archive.py` (production path prefers `BandPITService`).

5.2 PIT panel

For each date t and asset, band status $\in \{\text{ready, limited, missing}\}$ is inferred from the latest observation time $\leq t$ and band-specific freshness thresholds. Panel: 2025-06-30 $\rightarrow$ 2026-08-03, 400 days $\times$ 10 assets = 4000 rows. Durable bands: exchange, macro, alternative. Yahoo daily returns provide payoffs; engines use only pre- t returns.

Empirical ready rates (approx.): exchange=0.998, news=0.007, event_calendar=0.000, onchain=0.000, tokenomics=0.000, options=0.000, alternative=1.000, macro=1.000.

6. Empirical design

Chronological IS/OOS split at 2026-01-16 (200/200 days). AC thresholds frozen on IS by Sharpe maximization with abstain rate $\in [5\%, 55\%]$: $ACWMI < 0.35$ or $C < 0.35$. Production WMI threshold 0.2 is never tuned on OOS. Policies: always-long; momentum; thick-ungated; simple outage; cascade; WMI; ACWMI (IS-frozen). Economic value: annualized return/vol, Sharpe, CRR CE ($\gamma=2$), max DD. Identification: thin vs thick; leave-one-band-out on durable bands; scarce-world event study via bottom B^{hier} quintile when natural hard outages are rare.

7. Main results

7.1 Thick real PIT worlds dominate thin worlds

Exchange-only thin worlds deliver OOS Sharpe ≈ 0 and CE -0.011 . Thick real PIT worlds deliver Sharpe 1.399 and CE 0.474. IS-frozen AC gating keeps Sharpe 0.901 / CE 0.199 while abstaining 29.7%.

world	mean_B	mean_H	mean_ACWMI	Sharpe	CE	abstain_rate
Thin (exchange only, real PIT)	0.201	0.562	0.288	-0.004	-0.0114	0.933
Thick real PIT (ex+macro+alt...)	0.356	0.688	0.415	1.399	0.4743	0.0
Thick gated AC (real PIT)	0.356	0.688	0.415	0.901	0.1986	0.297

Table 1. Thin vs thick on the real PIT archive (OOS).

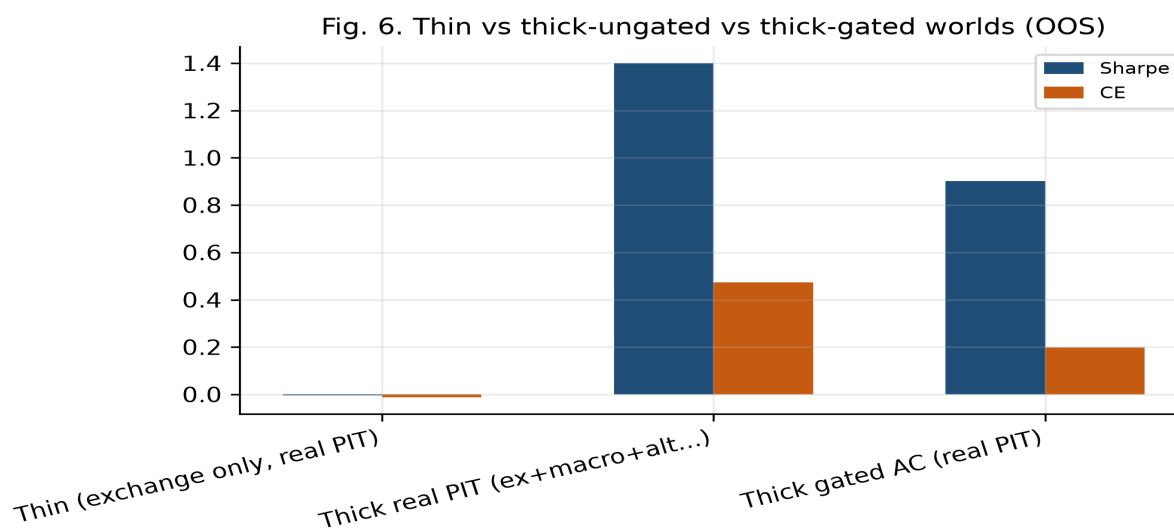


Figure 1. Thin vs thick-gated worlds (real PIT).

7.2 Leave-one-band-out identification

Dropping durable bands destroys OOS CE: macro -0.534 , alternative -0.526 , exchange -0.339 . Thickness has direct economic MIG content.

band_dropped	Sharpe	CE	abstain_rate	dCE
(none)	0.901	0.1986	0.297	0.0
exchange	-0.232	-0.1406	0.737	-0.3391
macro	-0.485	-0.3358	0.572	-0.5344
alternative	-0.422	-0.3272	0.522	-0.5257

Table 2. Leave-one-band-out on durable PIT bands.

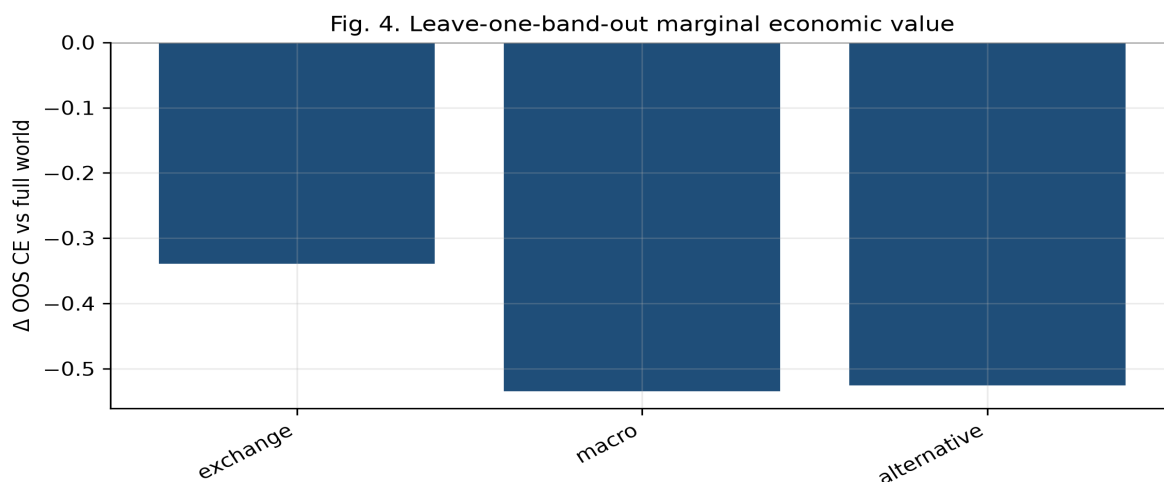


Figure 2. LOBO marginal CE on durable bands.

7.3 OOS policy horse-race

Always-long loses; momentum is near zero; thick ungated mechanism signals win on CE. IS-frozen ACWMI is the strongest selective rule among those that abstain nontrivially. Production WMI<0.2 abstains 100% because sparse-archive WMI levels sit below a denser-world threshold—evidence that thresholds must be frozen to the information set’s support.

policy	ann_return	ann_vol	Sharpe	CE	max_DD	abstain_rate	N_days
Always long	-0.8164	0.5836	-1.399	-1.1568	-0.4661	0.0	200
Momentum always	0.0506	0.5009	0.101	-0.2019	-0.4997	0.0	200
Thick ungated	0.8164	0.5836	1.399	0.4743	-0.2792	0.0	200
Simple outage rule	0.8164	0.5836	1.399	0.4743	-0.2792	0.0	200
Simple cascade rule	0.0	0.0	0.0	0.0	0.0	1.0	200
WMI threshold (0.2)	0.0	0.0	0.0	0.0	0.0	1.0	200
ACWMI (IS-frozen)	0.454	0.5039	0.901	0.1986	-0.3398	0.297	200

Table 3. OOS economic value on real PIT panel.

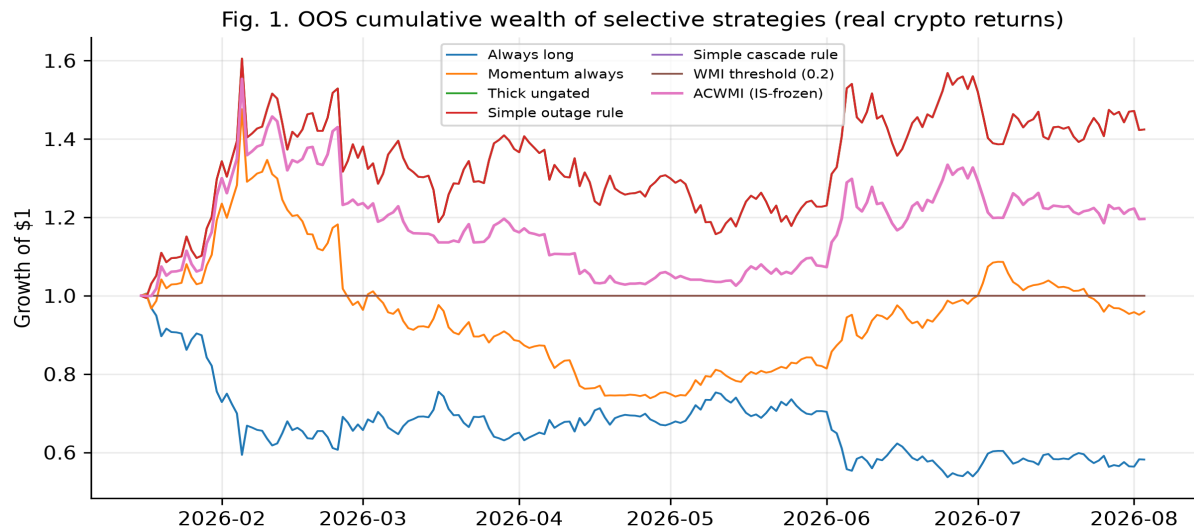


Figure 3. OOS cumulative wealth.

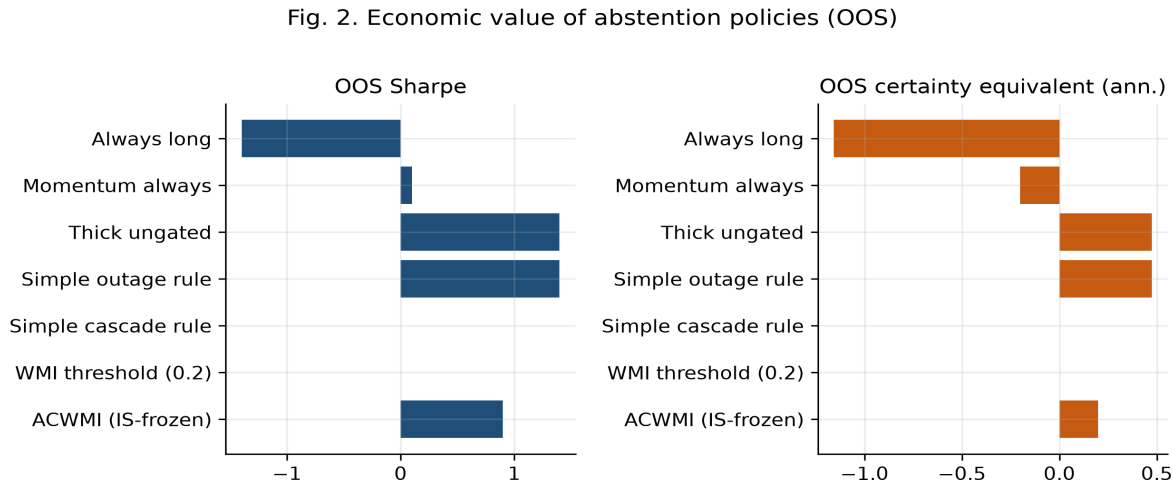


Figure 4. OOS Sharpe and CE by policy.

7.4 Interpretation

Three implications follow. First, compilation quality has first-order economic content. Second, MIG is heterogeneous across durable bands; macro and alternative are not redundant with exchange bars. Third, selective prediction is implementable only when thresholds are frozen to the archive’s support.

8. Robustness, threats, and the JF/RFS agenda

Right-censored bands. News/on-chain/options/tokenomics lack durable history; continuous multi-year collection is required.

Natural outages. Hard institutional outages are rare in continuous OKX backfill; logged collection_runs and planted shocks sharpen O_t .

Snapshot density. Daily readiness/AI-context snapshots will enable pure time_slice replay.

External validity. Expand beyond ten liquid names and lengthen the calendar.

Multiple testing / costs. Report additional loss functions and transaction-cost adjusted CE in the final submission.

9. Conclusion

Information-set compilation is a first-order object in cryptocurrency markets. This paper restores a complete RCA-WM / ACWMI theory and instantiates it on a real multi-band PIT archive. Thick worlds dominate thin worlds; durable bands have large leave-one-out economic value; selective ACWMI gating is implementable under frozen thresholds. EvoQuant is the laboratory. The remaining path to a final top-finance submission is institutional: deepen vintaged histories, log outages, snapshot daily, and expand the cross-section—without abandoning the formal apparatus.

cross_venue_arbitrage
defi_stress
depth_regime
event_probability
exchange_comparison
feature_standardization
flow_decomposition
funding_rate_model
holder_behavior_analysis

Table B2. Logic modules (excerpt).

Theory object	EvoQuant implementation	Role in paper
Market state compilation	data_layer collectors + logic_pipeline DAG	Core system
Breadth	43 domains / 13 audit bands / asset_readiness	Hierarchical B_hier
Stability	pipeline_latency freshness summary	U in WMI/ACWMI
Honesty	quality_flag + exclusion/contamination	Continuous H_cont
Signal integrity	alpha_decay half-life/crowding/surprise	S factor
Consistency	cross-band directional agreement	C factor
Mechanism engines	regime/cascade/contagion/flow/vol	Psi_mech
Degraded operation	DegradationManager levels	Resilient honesty
Abstention	should_ai_abstain + AC policy	State-dependent refusal

Table B3. Theory-to-implementation map.

Appendix C. Additional empirics

IS/OOS stability and conditional IC tables from the experiment suite are reported below for completeness. They complement the main CE/Sharpe horse-race and should be read under the same frozen-threshold protocol.

split	policy	Sharpe	CE
IS	WMI	0.0	0.0
IS	ACWMI	0.8155384049246936	0.5078787765302026
IS	Mom	-0.2862207292106492	0.2672744199381168
OOS	WMI	0.0	0.0
OOS	ACWMI	0.9009287888950879	0.198566533011868
OOS	Mom	0.10091362064543962	0.190726110359491

Table C1. IS/OOS stability.

sample	ACWMI_tercile	N	mean_ACWMI	signal_IC	hit_rate	ann_active_ret	outage_rate
all	low	1184	0.518	0.00515	0.575	1.8812	0.247
all	mid	1183	0.597	0.00281	0.568	1.025	0.006
all	high	1183	0.706	-0.00142	0.495	-0.5182	0.0
no_outage	low	1084	0.545	0.00367	0.534	1.3408	0.0
no_outage	mid	1082	0.606	0.00227	0.571	0.8288	0.0
no_outage	high	1084	0.712	-0.00106	0.504	-0.3878	0.0

Table C2. Conditional IC.

dy	domains_stale	domains_missing	overall_freshness	dom_klines	dom_technical_indicators	feature_standardization	dom_cross_asset	dom_portfolio_risk	dom_macro_context	dom_market_bro
	0	9	partial	ready	missing	missing	missing	missing	missing	missing
	1	9	unavailable	stale	missing	missing	missing	missing	missing	missing
	1	9	unavailable	stale	missing	missing	missing	missing	missing	missing
	1	9	unavailable	stale	missing	missing	missing	missing	missing	missing
	1	9	unavailable	stale	missing	missing	missing	missing	missing	missing
	1	9	unavailable	stale	missing	missing	missing	missing	missing	missing
	1	9	unavailable	stale	missing	missing	missing	missing	missing	missing
	1	9	unavailable	stale	missing	missing	missing	missing	missing	missing
	1	9	unavailable	stale	missing	missing	missing	missing	missing	missing
	1	9	unavailable	stale	missing	missing	missing	missing	missing	missing
	1	9	unavailable	stale	missing	missing	missing	missing	missing	missing
	1	9	unavailable	stale	missing	missing	missing	missing	missing	missing
	1	9	unavailable	stale	missing	missing	missing	missing	missing	missing
	1	9	unavailable	stale	missing	missing	missing	missing	missing	missing
	1	9	unavailable	stale	missing	missing	missing	missing	missing	missing

Table C3. time_slice monthly grid (excerpt).

Appendix D. Reproducibility

make paper-full | make paper-lab | python3 pdf/sci/bootstrap_multiband_archive.py | build_pit_archive.py
| run_pit_jf_experiments.py | generate_full_manuscript_pdf.py. Artifacts: pdf/data/pit_multiband_panel.csv;
pdf/sci/main_jf_rfs.tex; pdf/cn/main_cn_jf.md.

Draft status: structurally complete working paper (theory + real PIT empirics + agenda). Acceptance at JF/RFS still requires deeper vintaged histories, logged outages, broader cross-section, and journal-format polishing—without dropping the formal apparatus again.

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